Digit. Signal Process. 162 (2025) 105191

Contents lists available at ScienceDirect

Digital Signal Processing

journal homepage: www.elsevier.com/locate/dsp

The windowed two-dimensional graph fractional Fourier transform

Yu-Chen Gan [a,b], Jian-Yi Chen [a,b], Bing-Zhao Li [a,b,*]

[a] School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China
[b] Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China

ARTICLE INFO

Keywords:
Graph signal processing
Graph Fourier transform
Cartesian product graphs
Fractional Fourier transform

ABSTRACT

In the vibrant landscape of image and signal processing, the research on multi-dimensional graph signals has been making remarkable strides. However, despite the progress, the in-depth exploration and analysis of graph signals defined on two-dimensional (2-D) Cartesian product graphs still present certain gaps and challenges. This paper presents a comprehensive investigation into the generalization of the windowed graph fractional Fourier transform (WGFRFT) in the context of 2-D Cartesian product graphs. Firstly, the 2-D WGFRFT and its inverse transform are meticulously derived and defined, accompanied by the proposal of a fast algorithm. Subsequently, through an experiment, the advantages of the 2-D WGFRFT over the WGFRFT in processing two-dimensional graph signals are verified. Moreover, the effectiveness of the fast algorithm is rigorously validated through vertex-frequency analysis. And lastly, based on the 2-D WGFRFT, a novel filter learning method is put forward, and its potential in image classification is demonstrated.

1. Introduction

In contemporary data analysis, numerous real-world datasets inherently possess graph structures. Social networks, sensor networks, transportation networks, biological networks, and brain connectivity networks are just a few illustrations of such data, which are interconnected through complex graphs [1–6]. Graph signal processing (GSP) has emerged as a potent mathematical framework for analyzing and processing data defined on irregular domains represented by graphs [7–9]. GSP systematically extends classical signal processing concepts to the graph domain and enriches fundamental operations like filtering, transformation, sampling, and interpolation to adapt to graph-defined signals [10–15]. This approach provides a unified theoretical framework that connects discrete and continuous domains. GSP has exhibited remarkable effectiveness in diverse fields, including machine learning, image processing, and network science.

In the realm of GSP, one of the core tools is the graph Fourier transform (GFT), which extends the classical Fourier analysis to graph structures [11]. Through the utilization of the Laplacian operator or the eigen-decomposition of the adjacency matrix, GFT converts graph signals into the graph frequency domain [11,16]. In this domain, the transformation coefficients disclose how signals are manifested in different frequency components throughout the graph. To extract the local features of graph signals effectively, fractional processing methods have been introduced into GSP [17,18]. The integration of fractional parameters enables flexible and multi-scale analysis. Relevant studies cover the graph fractional Fourier transform (GFRFT) based on the graph adjacency matrix and the spectral graph fractional Fourier transform (SGFRFT) based on the graph Laplacian matrix [19,20]. These transformations have demonstrated advantages in emphasizing the local features of graph signals. Nevertheless, both GFRFT and SGFRFT, similar to GFT, possess a common limitation when it comes to presenting the spectra of multi-dimensional graph signals. Specifically, they neglect the product structure within graphs, thereby leading to insufficient characterization.

In the domain of GSP, Cartesian product graphs serve as an effective means to represent multi-domain data. Consequently, numerous signal processing methods predicated on Cartesian product graphs have been put forward [21–24]. With the aim of further tackling the directional traits of multi-dimensional graph signals and curtailing the computational complexity within the graph fractional domain, the multi-dimensional graph fractional Fourier transform (MGFRFT) has been devised [20]. MGFRFT utilizes Cartesian product graphs to depict graphs of elevated dimensionality and more intricate structures. By decomposing the fundamental graph into two or more factor graphs possessing fewer nodes, it extends the capabilities of GFRFT and SGFRFT. This decomposition notably diminishes the algorithmic complexity related to

* Corresponding author at: School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China.
E-mail address: li_bingzhao@bit.edu.cn (B.-Z. Li).

https://doi.org/10.1016/j.dsp.2025.105191

Available online 25 March 2025

the eigen-decomposition of GFRFT and SGFRFT. However, challenges remain in comprehensively characterizing local information.

Window functions play a pivotal role in signal processing, especially in the aspects of frequency domain analysis and filtering [25–30]. Through their shapes and characteristics, they are capable of customizing the performance of signals in both the frequency and time domains, thereby adapting to diverse signal processing tasks. In the domain of GSP, window functions are also utilized to explore the localized feature information of graph signals [31–34]. In this paper, we propose a novel transformation by integrating window functions into the two-dimensional (2-D) graph fractional Fourier transform [35]. The appropriate selection of window functions is of utmost importance in practical applications. This study delves into the design of window functions within the context of real-world image classification tasks.

This paper is structured as follows. In Section 2, the fundamental theories of signal processing and the graph Fourier transform are introduced. Subsequently, in Section 3, 2-D WGFRFT and its fast algorithm are defined. Moreover, the efficiency of the fast algorithm is demonstrated. Section 4 applies the 2-D WGFRFT to image classification tasks, thereby showcasing its potential in practical applications. Finally, Section 5 presents the conclusions of the paper.

2. Preliminaries

2.1. The graph signal

Graph signals are defined by mapping real (or complex) values to each vertex. In essence, they can be regarded as functions defined on a set of graph vertices, denoted as $\mathbf{f}: V \to \mathbb{C}$. For graphs with a finite number of vertices, assuming the number of vertices is N, the graph signal can be represented in the form of a vector:

$$\mathbf{f} = [f(1), f(2), \ldots, f(N)]^T. \tag{1}$$

2.2. Graph fractional Fourier transform

Assume that $G = (V, E, W)$ represents an undirected graph with N vertices. Here, W denotes the adjacency matrix of G, and $\mathcal{L} = D - W$ is defined as the graph Laplacian of G, where D is the diagonal degree matrix of G. Suppose that $\alpha \in [0, 1]$ is a fractional order. Let $\mathcal{L} = \mathbf{U}\Lambda\mathbf{U}^T$ be the Eigenvalue decomposition of $\mathcal{L}$, where $\mathbf{U}$ is an orthogonal matrix. Additionally, consider the eigenvalue decomposition of $\mathbf{U}$ given by $\mathbf{U} = \mathbf{P}\boldsymbol{\Xi}\mathbf{P}^H$. Then we have

$$\mathbf{U}^\alpha = \mathbf{P}\boldsymbol{\Xi}^\alpha\mathbf{P}^H. \tag{2}$$

The graph fractional Laplacian operator $\mathcal{L}_\alpha$ is defined by

$$\mathcal{L}_\alpha := \mathbf{U}^\alpha \Lambda^\alpha (\mathbf{U}^\alpha)^H = \boldsymbol{\kappa}\mathbf{R}\boldsymbol{\kappa}^H. \tag{3}$$

Note that

$$\boldsymbol{\kappa} = \left[\, \boldsymbol{\kappa}_1,\ \boldsymbol{\kappa}_2,\ \cdots,\ \boldsymbol{\kappa}_{N_i} \,\right] = \mathbf{U}^\alpha, \tag{4}$$

$$\mathbf{R} = \operatorname{diag}\left\{ r_1, r_2, \cdots, r_{N_i} \right\} = \Lambda^\alpha. \tag{5}$$

Let $\mathbf{f}: V \to \mathbb{C}$ be a graph signal. Then SGFRFT can be defined by

$$\widehat{\mathbf{f}}_\alpha = \kappa^H \mathbf{f}, \tag{6}$$

with another form of

$$\widehat{f}_\alpha(l) = \langle \mathbf{f}, \boldsymbol{\kappa}_l \rangle = \sum_{n=1}^{N} f(n)\kappa_l^*(n). \tag{7}$$

The inverse SGFRFT is given by

$$\mathbf{f} = \kappa \widehat{\mathbf{f}}_\alpha. \tag{8}$$

2.3. Cartesian product graph

Consider two graphs $G_1 = (V_1, E_1, W_1)$ and $G_2 = (V_2, E_2, W_2)$ with N_1 and N_2 vectors respectively. The Cartesian product graph of G_1 and G_2 are given by $G_1 \square G_2$ with vertex set $V_1 \times V_2$. The edge set $E_\square$ of $G_1 \square G_2$ satisfies:

$$\left((i_1, i_2), (k_1, k_2)\right) \in E_\square \tag{9}$$

$$\Longleftrightarrow \left[(i_1, k_1) \in E_1, i_2 = k_2\right] \quad \text{or} \quad \left[i_1 = k_1, (i_2, k_2) \in E_2\right], \tag{10}$$

and the weight function $W_\square$ is given by:

$$W_\square\left((i_1, i_2), (j_1, j_2)\right) = W_1(i_1, j_1)\,\delta(i_2, j_2) + \delta(i_1, j_1)\,W_2(i_2, j_2), \tag{11}$$

where δ is the delta function.

Based on this definition, there are some obvious properties of Cartesian product graphs:

1. associative: the graphs $(F \square G) \square H$ and $F \square (G \square H)$ are naturally isomorphic.
2. commutative: the graphs $G \square H$ and $H \square G$ are naturally isomorphic.

Graphs G_1 and G_2 are called factor graphs of $G_1 \square G_2$. The adjacency matrix $W_1 \oplus W_2$ and Laplacian matrix $\mathcal{L}_1 \oplus \mathcal{L}_2$ of $G_1 \square G_2$ are computed by:

$$W_1 \oplus W_2 = W_1 \otimes I_{N_2} + I_{N_1} \otimes W_2, \tag{12}$$

$$\mathcal{L}_1 \oplus \mathcal{L}_2 = \mathcal{L}_1 \otimes I_{N_2} + I_{N_1} \otimes \mathcal{L}_2. \tag{13}$$

Here, the operator $\oplus$ represents the Kronecker sum and $\otimes$ represents the Kronecker product [36]. If A is a $m \times n$ matrix and B is a $p \times q$ matrix, then the Kronecker product $A \otimes B$ is the $pm \times qn$ block matrix:

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}. \tag{14}$$

2.4. Laplacian-based 2-D graph fractional Fourier transform

Consider a Cartesian product $G_1 \square G_2$ of graphs $G_1 = (V_1, E_1, W_1)$ and $G_2 = (V_2, E_2, W_2)$. The Eigenvalue decomposition of the Laplacian matrices of factor graphs G_1 and G_2 are defined as $\mathcal{L}_1 = \mathbf{U}_1\Lambda_1\mathbf{U}_1^T$ and $\mathcal{L}_2 = \mathbf{U}_2\Lambda_2\mathbf{U}_2^T$. The corresponding graph fractional Laplacian operators of $\mathcal{L}_1$ and $\mathcal{L}_2$ are

$$\mathcal{L}_\alpha^{(i)} = \boldsymbol{\kappa}^{(i)}\mathbf{R}^{(i)}(\boldsymbol{\kappa}^{(i)})^H, \tag{15}$$

where

$$\boldsymbol{\kappa}^{(i)} = \left[\, \boldsymbol{\kappa}_1^{(i)},\ \boldsymbol{\kappa}_2^{(i)},\ \cdots,\ \boldsymbol{\kappa}_{N_i}^{(i)} \,\right] = \mathbf{U}_i^\alpha, \tag{16}$$

$$\mathbf{R}^{(i)} = \operatorname{diag}\left\{ r_1^{(i)}, r_2^{(i)}, \cdots, r_{N_i}^{(i)} \right\} = \Lambda_i^\alpha, \tag{17}$$

with i = 1,2.

Therefore, for $G_1 \square G_2$, we have the following eigenvalue decomposition of $\mathcal{L}_\alpha^{(1)} \oplus \mathcal{L}_\alpha^{(2)}$ in graph fractional domain:

$$\left(\mathcal{L}_\alpha^{(1)} \oplus \mathcal{L}_\alpha^{(2)}\right) \begin{pmatrix} \kappa_{l_1}^{(1)}(1)\kappa_{l_2}^{(2)}(1) \\ \kappa_{l_1}^{(1)}(1)\kappa_{l_2}^{(2)}(2) \\ \vdots \\ \kappa_{l_1}^{(1)}(N_1)\,\kappa_{l_2}^{(2)}(N_2) \end{pmatrix}$$

$$= \left(r_{l_1}^{(1)} + r_{l_2}^{(2)}\right) \begin{pmatrix} \kappa_{l_1}^{(1)}(1)\kappa_{l_2}^{(2)}(1) \\ \kappa_{l_1}^{(1)}(1)\kappa_{l_2}^{(2)}(2) \\ \vdots \\ \kappa_{l_1}^{(1)}\left(N_1\right)\kappa_{l_2}^{(2)}\left(N_2\right) \end{pmatrix}. \tag{18}$$

Therefore, The Laplacian-based 2-D graph fractional Fourier transform of a 2-D signal $\mathbf{f} : V_1 \times V_2 \to \mathbb{C}$ on a Cartesian product graph $G_1 \square G_2$ is

$$\hat{f}_\alpha\left(l_1 + l_2\right) = \sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2} f\left(n_1, n_2\right)\left(\kappa_{l_1}^{(1)}\left(n_1\right)\right)^*\left(\kappa_{l_2}^{(2)}\left(n_2\right)\right)^*, \tag{19}$$

for $l_1 = 1, 2, \cdots, N_1$, $l_2 = 1, 2, \cdots, N_2$. And its inverse is given by

$$f\left(n_1, n_2\right) = \sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2} \hat{f}_\alpha\left(l_1 + l_2\right)\kappa_{l_1}^{(1)}\left(n_1\right)\kappa_{l_2}^{(2)}\left(n_2\right), \tag{20}$$

for $n_1 = 1, 2, \cdots, N_1$, $n_2 = 1, 2, \cdots, N_2$.

2.5. WGFRFT

Assume that $G = (V, E)$ is an undirected graph with N vertices. The basics of SGFRFT are given by κ_l. Given a window function $\psi : V \to \mathbb{C}$, the WGFRFT of signal $\mathbf{f}$ on graph G is given by [32]

$$\left(W_{\psi,\alpha}\mathbf{f}\right)(i,l) = N^\alpha \sum_{d=1}^{N} f(d)\left(\sum_{m=1}^{N}\hat{\varphi}_\alpha^*(m)\kappa_m(i)\kappa_m^*(d)\right)\kappa_l^*(d). \tag{21}$$

Let

$$\begin{aligned}\boldsymbol{\Psi}_{i,l}(n) &= (M_l^\alpha T_i^\alpha \psi)(n) \\ &= N^\alpha \kappa_l(n)\sum_{m=1}^{N}\hat{\varphi}_\alpha(m)\kappa_m^*(i)\kappa_m(n).\end{aligned} \tag{22}$$

Then function family $\left\{\boldsymbol{\Psi}_{i,l}\right\}_{i=1,2,\cdots,N;\ l=1,2,\cdots,N}$ form a framework. Therefore

$$\left(W_{\psi,\alpha}\mathbf{f}\right)(i,l) = \left\langle \mathbf{f}, \boldsymbol{\Psi}_{i,l}\right\rangle. \tag{23}$$

When $\alpha = 1$, the WGFRFT degenerates into the windowed graph Fourier transform (WGFT) defined by Shuman [31]. And the inverse of WGFRFT is given by

$$f(n) = \frac{1}{N^\alpha \left\|T_i^\alpha \boldsymbol{\psi}\right\|_2^2}\sum_{i=1}^{N}\sum_{l=1}^{N}\left(W_{\psi,\alpha}\mathbf{f}\right)(i,l)\boldsymbol{\Psi}_{i,l}(n), \tag{24}$$

where $n \in \{1, 2, \cdots, N\}$.

3. The 2-D WGFRFT

In this section, we present the theoretical foundations and formulations of two fundamental operators: the 2-D graph fractional translation operator and the 2-D graph fractional modulation operator. Building upon these operators, we establish the mathematical framework for the 2-D WGFRFT and formally derive its inverse transformation. Furthermore, we propose an efficient fast algorithm that significantly reduces the computational complexity while maintaining analytical accuracy. To demonstrate the practical utility and computational benefits of our approach, we implement both the standard 2-D WGFRFT and its fast algorithm through two examples, providing validation of the theoretical framework.

3.1. Definition

Consider a Cartesian product $G_\square = G_1 \square G_2$ of graphs $G_1 = (V_1, E_1, W_1)$ and $G_2 = (V_2, E_2, W_2)$. The eigenvalue decomposition of the Laplacian matrices of factor graphs G_1 and G_2 are defined as $\mathcal{L}_1 = \mathbf{U}_1\Lambda_1\mathbf{U}_1^T$ and $\mathcal{L}_2 = \mathbf{U}_2\Lambda_2\mathbf{U}_2^T$. The corresponding graph fractional Laplacian operators of $\mathcal{L}_1$ and $\mathcal{L}_2$ are

$$\mathcal{L}_\alpha^{(i)} = \boldsymbol{\kappa}^{(i)}\mathbf{R}^{(i)}(\boldsymbol{\kappa}^{(i)})^H, \tag{25}$$

where

$$\boldsymbol{\kappa}^{(i)} = \left[\boldsymbol{\kappa}_1^{(i)},\ \boldsymbol{\kappa}_2^{(i)},\ \cdots,\ \boldsymbol{\kappa}_{N_i}^{(i)}\right] = \mathbf{U}_i^\alpha, \tag{26}$$

$$\mathbf{R}^{(i)} = \mathrm{diag}\left\{r_1^{(i)}, r_2^{(i)}, \cdots, r_{N_i}^{(i)}\right\} = \Lambda_i^\alpha, \tag{27}$$

with i = 1,2. Thus the graph fractional Laplacian operator of $G_\square$ is

$$\mathcal{L}_\alpha^{prod} = \mathcal{L}_\alpha^{(1)} \oplus \mathcal{L}_\alpha^{(2)} = \boldsymbol{\kappa}^{prod}\mathbf{R}^{prod}(\boldsymbol{\kappa}^{prod})^H. \tag{28}$$

And we have

$$\mathcal{L}_\alpha^{prod}\begin{pmatrix} \kappa_{l_1}^{(1)}(1)\kappa_{l_2}^{(2)}(1) \\ \kappa_{l_1}^{(1)}(1)\kappa_{l_2}^{(2)}(2) \\ \vdots \\ \kappa_{l_1}^{(1)}\left(N_1\right)\kappa_{l_2}^{(2)}\left(N_2\right) \end{pmatrix} = \left(r_{l_1}^{(1)} + r_{l_2}^{(2)}\right)\begin{pmatrix} \kappa_{l_1}^{(1)}(1)\kappa_{l_2}^{(2)}(1) \\ \kappa_{l_1}^{(1)}(1)\kappa_{l_2}^{(2)}(2) \\ \vdots \\ \kappa_{l_1}^{(1)}\left(N_1\right)\kappa_{l_2}^{(2)}\left(N_2\right) \end{pmatrix}. \tag{29}$$

For any 2-D signals $\mathbf{f}, \mathbf{g} : V_1 \times V_2 \to \mathbb{C}$ on a Cartesian product graph $G_1 \square G_2$, the 2-D graph fractional convolution operator $*_\alpha$ is defined by

$$\left(\mathbf{f} *_\alpha \mathbf{g}\right)(n_1, n_2) = \sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\hat{f}_\alpha(l_1, l_2)\hat{g}_\alpha(l_1, l_2)\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2). \tag{30}$$

In the definition of the classical short-time fractional Fourier transform, there exists a more intuitive interpretation [37]. That is, the translation of the window function is realized by the translation operator. Subsequently, the fractional Fourier transform is carried out to yield the result. By leveraging the concept of the short-time fractional Fourier transform, the definition of the windowed graph fractional Fourier transform requires the introduction of new translation and modulation operators.

For the purpose of moving the window function across the vertex domain, we utilize the 2-D graph fractional convolution of delta function δ_{i_1,i_2}, which is centered on vertex (i_1, i_2) along with the graph signal $\mathbf{f}$ to define the graph fractional translation operator T_{i_1,i_2}^α.

Definition 1. For any 2-D signal $\mathbf{f}$ on graph $G_1 \square G_2$ and any vector (i_1, i_2) in graph $G_1 \square G_2$, the 2-D graph fractional translation operator is defined by

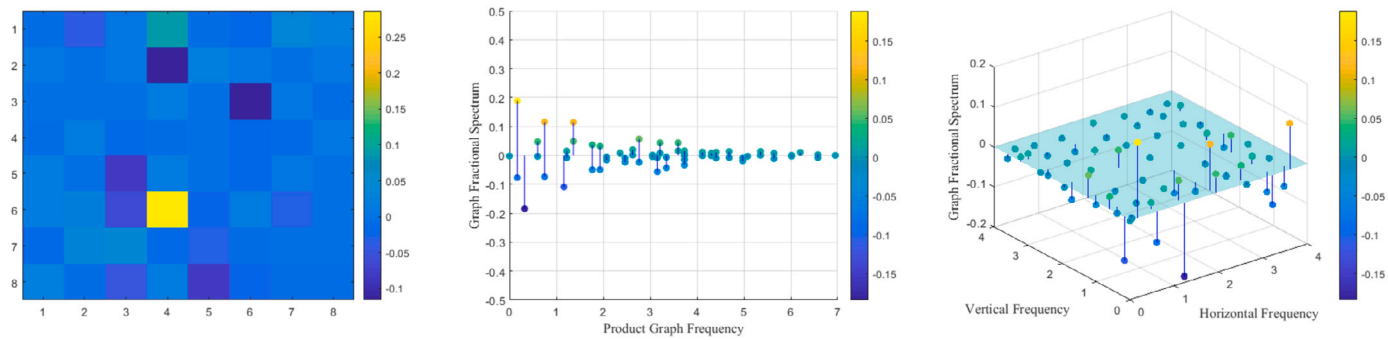
$$\begin{aligned}&\left(T_{i_1,i_2}^\alpha \mathbf{f}\right)(n_1, n_2) \\ &= (\sqrt{N_1 N_2})^\alpha\left(\mathbf{f} *_\alpha \boldsymbol{\delta}_{i_1,i_2}\right)(n_1, n_2) \\ &= (\sqrt{N_1 N_2})^\alpha \sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\hat{f}_\alpha(l_1, l_2)\hat{\delta}_\alpha(l_1, l_2)\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2) \\ &= (\sqrt{N_1 N_2})^\alpha \sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\hat{f}_\alpha(l_1, l_2)\kappa_{l_1}^{(1)*}(i_1)\kappa_{l_2}^{(2)*}(i_2)\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2),\end{aligned} \tag{31}$$

where $\boldsymbol{\delta}_{i_1,i_2}$ is a 2-D matrix with 1 at the (i_1, i_2) position and 0 at the rest positions.

Thus the 2-D graph fractional modulation operator is defined.

Definition 2. For any 2-D signal $\mathbf{f}$ on graph $G_1 \square G_2$ and any $l_1 \in \left\{1, 2, \ldots, N_1\right\}$, $l_2 \in \left\{1, 2, \ldots, N_2\right\}$, the 2-D graph fractional modulation operator M_{l_1,l_2}^α is defined by

$$\left(M_{l_1,l_2}^\alpha \mathbf{f}\right)(n) = (\sqrt{N_1 N_2})^\alpha f(n_1, n_2)\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2). \tag{32}$$



(a) filtered signal (b) 1-D spectrum (c) 2-D spectrum

Fig. 1. The filtered time-domain signal and its spectra.

Consequently, based on the preceding theoretical foundations, the 2-D graph fractional modulation operator is formally and precisely defined.

Definition 3. As a window function $\psi : V_1 \times V_2 \to \mathbb{C}$ is given, the 2-D WGFRFT of 2-D signal $\mathbf{f}$ on graph $G_1 \square G_2$ is defined by

$$\begin{aligned}\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1,l_2) =& (N_1N_2)^{\alpha}\sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2} f(n_1,n_2)\\ &\times\Bigg(\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\hat{\varphi}_{\alpha}^{*}(m_1,m_2)\kappa_{m_1}^{(1)}(i_1)\kappa_{m_2}^{(2)}(i_2)\\ &\times\kappa_{m_1}^{(1)*}(n_1)\kappa_{m_2}^{(2)*}(n_2)\Bigg)\kappa_{l_1}^{(1)*}(n_1)\kappa_{l_2}^{(2)*}(n_2).\end{aligned} \tag{33}$$

Let

$$\begin{aligned}\Psi_{i_1,i_2,l_1,l_2}(n_1,n_2) =& \left(M_{l_1,l_2}^{\alpha}T_{i_1,i_2}^{\alpha}(\psi)\right)(n_1,n_2)\\ =& (N_1N_2)^{\alpha}\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2)\\ &\times\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\hat{\varphi}_{\alpha}(m_1,m_2)\kappa_{m_1}^{(1)*}(i_1)\kappa_{m_2}^{(2)*}(i_2)\kappa_{m_1}^{(1)}(n_1)\kappa_{m_2}^{(2)}(n_2),\end{aligned} \tag{34}$$

thus

$$\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1,l_2) = \left\langle \mathbf{f}, \Psi_{i_1,i_2,l_1,l_2}\right\rangle. \tag{35}$$

In this case, we compose the 2-D graph fractional modulation operator and the 2-D graph fractional translation operator and apply them to the window function ψ. Then, the 2-D WGFRFT of $\mathbf{f}$ can be regarded as the convolution of the signal $\mathbf{f}$ and the result mentioned above.

There's a more intuitive elucidation of the 2-D WGFRFT. Specifically, if we set $h_{i_1,i_2}(n_1,n_2) = (\sqrt{N_1N_2})^{\alpha}\left(\mathbf{f}\circ\left(T_{i_1,i_2}^{\alpha}\psi\right)^{*}\right)(n_1,n_2)$, it then represents

$$\begin{aligned}\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1,l_2) =& \sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2} h_{i_1,i_2}(n_1,n_2)\kappa_{l_1}^{(1)*}(n_1)\kappa_{l_2}^{(2)*}(n_2)\\ =& (\widehat{h_{i_1,i_2}})_{\alpha}(l_1,l_2).\end{aligned} \tag{36}$$

In this case, $\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,-)$ is equal to the Laplacian-based 2-D graph fractional Fourier transform of $\mathbf{h_{i_1,i_2}}$.

Having delved into the details and significance of the 2-D WGFRFT, it becomes imperative to explore its counterpart: the inverse transform.

Theorem 1. *If the window function $\psi : V_1 \times V_2 \to \mathbb{C}$ satisfies $T_{i_1,i_2}^{\alpha}(\psi) \neq 0$, for any 2-D signal $\mathbf{f}$, the inverse windowed 2-D graph fractional Fourier transform (I-2-D WGFFT) is given by*

$$\begin{aligned}f(n_1,n_2) =& \frac{1}{(N_1N_2)^{\alpha}\left\|T_{i_1,i_2}^{\alpha}\psi\right\|_2^2}\\ &\times\sum_{i_1=1}^{N_1}\sum_{i_2=1}^{N_2}\sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1,l_2)\Psi_{i_1,i_2,l_1,l_2}(n_1,n_2).\end{aligned} \tag{37}$$

The proof is provided in Appendix A.

An essential element of our proposed algorithm lies in the window function. Within the framework of our algorithm, the window function functions in such a way that a sliding window traverses each vertex, serving as an effective frequency-domain filter. To provide an intuitive illustration of the window function's significance, we visualize the time-domain and frequency-domain signals post-filtering at the moment when the sliding window reaches one particular vertex in Example 1.

Example 1. Let us consider two path graphs, denoted as G_1 and G_2, with each having a length of 8. Subsequently, we define $G_{\square} = G_1 \square G_2$ as the Cartesian product graph formed by G_1 and G_2. It should be noted that graph $G_{\square}$ then encompasses a total of 64 vertices. Furthermore, we take into account a signal whose length is 64 and which is defined on graph $G_{\square}$, which is

$$\mathbf{f} = [0,\dots,0,1,0,\dots,0]. \tag{38}$$

The graph signal $\mathbf{f}$ assumes a value of 1 at the 30th index, while it takes a value of 0 at all other positions. We proceed to apply the WGFRFT and the 2-D WGFRFT to the signal $\mathbf{f}$. By doing so, we can extract both the one-dimensional (1-D) and 2-D spectra. Subsequently, the inverse transform is carried out to retrieve the time-domain signal plot after the filtering process. The outcomes of these operations are visually presented in Fig. 1.

Upon examination, it becomes evident that the filtered time-domain signal exhibits a comparatively large magnitude at the position $(4,6)$. In both the 1-D and 2-D spectra, the spectral coefficients manifest larger values within the low-frequency domain. More precisely, in the case of the 2-D spectrum, the coefficients possess larger magnitudes when the vertical frequency falls within the low-frequency interval. Notably, spectral coefficients obtained through the 1-D transformation of 2-D graph signals commonly display multi-valued traits at particular graph frequencies. By contrast, such equivocalness is intrinsically mitigated in the 2-D transformation. Consequently, the 2-D WGFRFT framework proffers a reliable remedy to the conundrum of multi-valued spectral coefficients.

3.2. Fast algorithm

Let us consider a Cartesian product graph $G_1 \square G_2$ with $N_1 \times N_2$ vertices. There are $(N_1 \times N_2)^2$ elements in $W_{\psi,\alpha}\mathbf{f}$. For each individual element in $W_{\psi,\alpha}\mathbf{f}$, the computational complexity is $O((N_1 \times N_2)^2)$. Consequently, the total computational complexity is $O((N_1 \times N_2)^4)$. To reduce this complexity, a fast calculation method is proposed.

Firstly, the $\mathbf{\Theta}$-field is defined by the Laplacian-based 2-D graph fractional Fourier transform of $W_{\psi,\alpha}\mathbf{f}$, which is

$$\Theta\left(l_1,l_2,l_1',l_2'\right) = \sum_{i_1=1}^{N_1}\sum_{i_2=1}^{N_2}\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1',l_2')\kappa_{l_1}^{(1)*}(i_1)\kappa_{l_2}^{(2)*}(i_2). \tag{39}$$

Moreover it can be written as

$$\begin{aligned}
&\Theta\left(l_1,l_2,l_1',l_2'\right)\\
=&\sum_{i_1=1}^{N_1}\sum_{i_2=1}^{N_2}\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1',l_2')\kappa_{l_1}^{(1)*}(i_1)\kappa_{l_2}^{(2)*}(i_2)\\
=&\sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2}\Bigg((N_1N_2)^{\alpha}\sum_{d_1=1}^{N_1}\sum_{d_2=1}^{N_2}f(d_1,d_2)\\
&\times\Big(\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\hat{\varphi}_{\alpha}^{*}(m_1,m_2)\kappa_{m_1}^{(1)}(n_1)\kappa_{m_2}^{(2)}(n_2)\\
&\times\kappa_{m_1}^{(1)*}(d_1)\kappa_{m_2}^{(2)*}(d_2)\Big)\kappa_{l_1'}^{(1)*}(d_1)\kappa_{l_2'}^{(2)*}(d_2)\Bigg)\kappa_{l_1}^{(1)*}(n_1)\kappa_{l_2}^{(2)*}(n_2)\\
=&(N_1N_2)^{\alpha}\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\sum_{d_1=1}^{N_1}\sum_{d_2=1}^{N_2}f(d_1,d_2)\kappa_{l_1'}^{(1)*}(d_1)\kappa_{l_2'}^{(2)*}(d_2)\\
&\times\kappa_{m_1}^{(1)*}(d_1)\kappa_{m_2}^{(2)*}(d_2)\hat{\varphi}_{\alpha}^{*}(m_1,m_2)\\
&\times\left(\sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2}\kappa_{m_1}^{(1)}(n_1)\kappa_{m_2}^{(2)}(n_2)\kappa_{l_1}^{(1)*}(n_1)\kappa_{l_2}^{(2)*}(n_2)\right)\\
=&(N_1N_2)^{\alpha}\sum_{d_1=1}^{N_1}\sum_{d_2=1}^{N_2}f(d_1,d_2)\kappa_{l_1'}^{(1)*}(d_1)\kappa_{l_2'}^{(2)*}(d_2)\\
&\times\kappa_{l_1}^{(1)*}(d_1)\kappa_{l_2}^{(2)*}(d_2)\hat{\varphi}_{\alpha}^{*}(l_1,l_2)\\
=&(N_1N_2)^{\alpha}\tilde{f}\left(l_1,l_2,l_1',l_2'\right)\hat{\varphi}_{\alpha}^{*}(l_1,l_2),
\end{aligned} \tag{40}$$

where

$$\tilde{f}\left(l_1,l_2,l_1',l_2'\right) = \sum_{d_1=1}^{N_1}\sum_{d_2=1}^{N_2}f(d_1,d_2)\kappa_{l_1'}^{(1)*}(d_1)\kappa_{l_2'}^{(2)*}(d_2)\kappa_{l_1}^{(1)*}(d_1)\kappa_{l_2}^{(2)*}(d_2). \tag{41}$$

Therefore, an efficient fast algorithm of 2-D WGFRFT is developed as follows.

Definition 4. The fast windowed 2-D graph fractional Fourier transform (2-D WFGFRFT) is defined by the inverse Laplacian-based 2-D graph fractional Fourier transform of $\mathbf{\Theta}$, which is

$$\begin{aligned}
\left(FW_{\psi,\alpha}\mathbf{f}\right)\left(n_1,n_2,l_1',l_2'\right) &= \sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\Theta\left(l_1,l_2,l_1',l_2'\right)\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2)\\
&=\sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}(N_1N_2)^{\alpha}\tilde{f}\left(l_1,l_2,l_1',l_2'\right)\hat{\varphi}_{\alpha}^{*}(l_1,l_2)\\
&\quad\times\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2).
\end{aligned} \tag{42}$$

We now proceed to analyze the computational complexity of the 2-D WFGFRFT.

1. First, we calculate

$$\begin{aligned}
&\tilde{f}\left(l_1,l_2,l_1',l_2'\right)\\
&=\sum_{d_1=1}^{N_1}\sum_{d_2=1}^{N_2}f(d_1,d_2)\kappa_{l_1'}^{(1)*}(d_1)\kappa_{l_2'}^{(2)*}(d_2)\kappa_{l_1}^{(1)*}(d_1)\kappa_{l_2}^{(2)*}(d_2).
\end{aligned} \tag{43}$$

   This computational step exhibits a complexity of $O((N_1 \times N_2)^3)$.
2. Subsequently, we expand $\hat{\varphi}_{\alpha}^{*}(l_1,l_2)$ as $\hat{\varphi}_{\alpha}^{*}(l_1,l_2,-,-)$ which is denoted as $\hat{\mathbf{\Psi}}_{\alpha}$ and $\tilde{f}\left(l_1,l_2,l_1',l_2'\right)$ can be expressed as $\tilde{\mathbf{F}}$. Then we calculate

$$\mathbf{\Theta} = (N_1N_2)^{\alpha}(\hat{\mathbf{\Psi}}_{\alpha})^{*}\cdot\tilde{\mathbf{F}}. \tag{44}$$

   The computational complexity of this step is equivalent to that of matrix multiplication, which is $O((N_1 \times N_2)^3)$.
3. Finally, we apply the inverse Laplacian-based 2-D graph fractional Fourier transform to $\mathbf{\Theta}$, giving by

$$FW_{\psi,\alpha}\mathbf{f} = \boldsymbol{\kappa}\cdot\mathbf{\Theta}. \tag{45}$$

   This computational step exhibits a complexity of $O((N_1 \times N_2)^3)$.

In summary, the computational complexity of the 2-D WFGFRFT is $O((N_1 \times N_2)^3)$. As the dimension N_1 and N_2 increase, this optimized approach demonstrates significantly enhanced computational efficiency compared to the original transformation, which exhibits a complexity of $O((N_1 \times N_2)^4)$.

In order to validate the effectiveness and superiority of the 2-D WGFRFT as well as its fast algorithm, we conduct vertex-frequency analysis on the signal defined over the product graph and subsequently plot the vertex-frequency image of the time-domain signals.

Example 2. Let us consider two path graphs G_1 and G_2, with each having a length of 15. Subsequently, we define $G_{\square} = G_1 \square G_2$ as the Cartesian product graph formed by G_1 and G_2. It should be noted that graph $G_{\square}$ then encompasses a total of 15×15 vertices. The graph fractional Laplacian operators corresponding to G_1 and G_2 are

$$\mathcal{L}_{\alpha}^{(i)} = \boldsymbol{\kappa}^{(i)}\mathbf{R}^{(i)}(\boldsymbol{\kappa}^{(i)})^{H}, \tag{46}$$

with i = 1, 2.

Let us take into account a 2-D graph signal $\mathbf{f}_1$ defined on this Cartesian product graph, which is

$$\mathbf{f}_1 = Re(\kappa_8^{(1)} \otimes \kappa_5^{(2)}). \tag{47}$$

Herein, $\kappa_8^{(1)}$ stands for the 8th column of the SGFRFT matrix corresponding to the first factor graph, while $\kappa_5^{(2)}$ represents the 5th column of the SGFRFT matrix associated with the second factor graph. The symbol $\otimes$ indicates the outer product, and it is elaborated as follows:

$$f_1(n_1,n_2) = Re\begin{pmatrix} \kappa_8^{(1)}(1)\kappa_5^{(2)}(1) & \kappa_8^{(1)}(1)\kappa_5^{(2)}(2) & \dots & \kappa_8^{(1)}(1)\kappa_5^{(2)}(15)\\ \kappa_8^{(1)}(2)\kappa_5^{(2)}(1) & \kappa_8^{(1)}(2)\kappa_5^{(2)}(2) & & \\ \vdots & & \ddots & \\ \kappa_8^{(1)}(15)\kappa_5^{(2)}(1) & & & \kappa_8^{(1)}(15)\kappa_5^{(2)}(15) \end{pmatrix}. \tag{48}$$

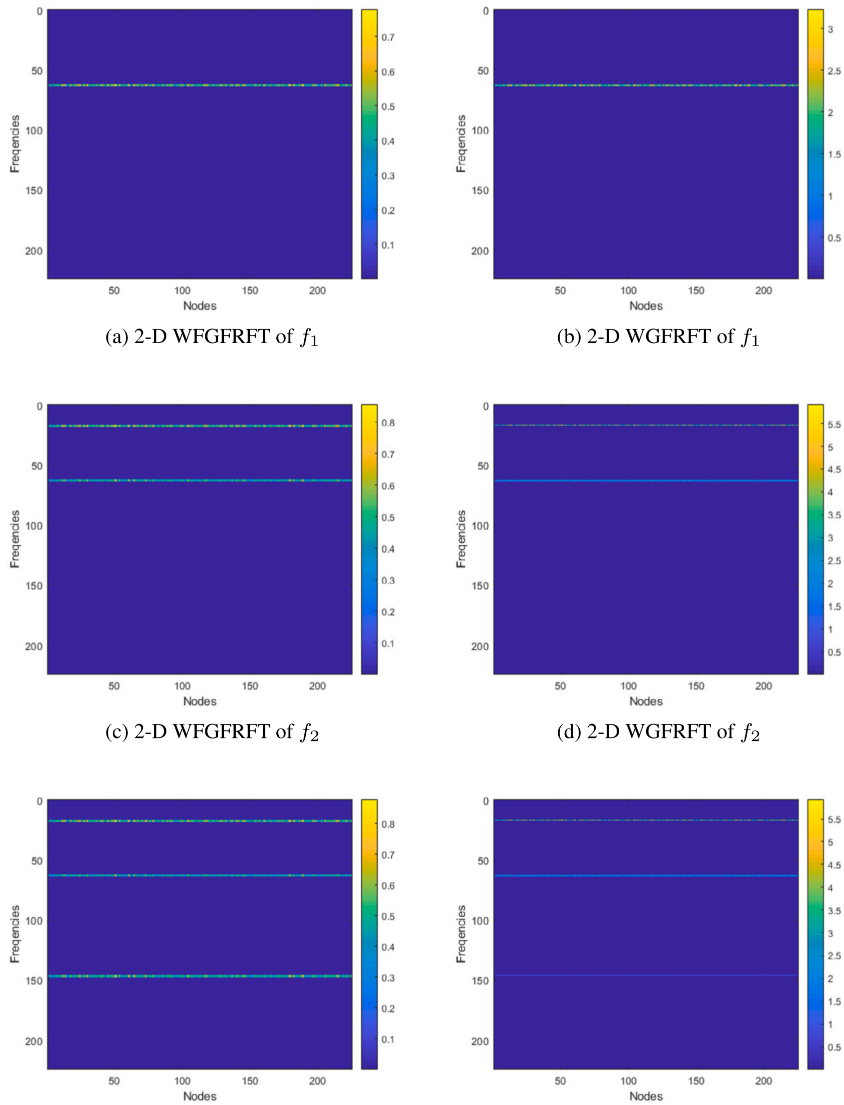
Similarly, we define two additional signals $\mathbf{f}_2$ and $\mathbf{f}_3$ by

$$\mathbf{f}_2 = Re(\kappa_8^{(1)} \otimes \kappa_5^{(2)} + \kappa_2^{(1)} \otimes \kappa_5^{(2)}), \tag{49}$$

$$\mathbf{f}_3 = Re(\kappa_8^{(1)} \otimes \kappa_5^{(2)} + \kappa_2^{(1)} \otimes \kappa_5^{(2)} + \kappa_{10}^{(1)} \otimes \kappa_8^{(2)}). \tag{50}$$

The 2-D WGFRFT and the 2-D WFGFRFT are applied to the aforementioned three signals. The experimental results are shown in Fig. 2.

Vertex-frequency image analysis offers robust validation of the alignment between experimental outcomes and theoretical predictions. Specifically, the vertex-frequency representations derived from both the standard 2-D WGFRFT and its optimized implementation exhibit a high degree of congruence, affirming the efficacy of our proposed fast algorithm. And it substantially reduces the computational complexity



(a) 2-D WFGFRFT of f_1 (b) 2-D WGFRFT of f_1

(c) 2-D WFGFRFT of f_2 (d) 2-D WGFRFT of f_2

(e) 2-D WFGFRFT of f_3 (f) 2-D WGFRFT of f_3

Fig. 2. Vertex-frequency Images of f_1 f_2 and f_3.

while preserving analytical precision. The consistency between these results substantiates the practical utility of our fast algorithm for vertex-frequency analysis.

4. Application

In this section, our focus turns to explore the applicability of 2-D WGFRFT within the realm of image classification. More precisely, we put forward a novel filter learning methodology predicated on 2-D WGFRFT and showcase its efficacy when applied to image classification undertakings.

In the simulation experiments, we make use of the MNIST dataset, which serves as a widely acknowledged benchmark within the domain of machine learning and computer vision [38–40]. The MNIST dataset, a benchmark for machine learning, consists of 70,000 grayscale handwritten digits (0-9) in 28×28 pixel format. Each image thereby provides 784 features. It is split into a 60,000-image training set and a 10,000-image test set. Renowned for its simplicity and high intra-class variability, MNIST is widely used to evaluate algorithms in image classification, feature extraction, and pattern recognition tasks.

4.1. *Experiment*

Given that differentiating between the digits 4 and 9 poses the greatest challenge within this dataset, we opted to select 1,000 samples respectively from the data labeled as 4 and 9. Fig. 3 showcases some of these samples.

There are 2000 samples, which are denoted as $N = 2000$. For the graph-based model, we take into account a Cartesian product graph $G_{\square}$ of two path graphs of length 28, which gives rise to a graph with $M = |V| = 784$ vertices. According to the relevant theories of the 2-D WGFRFT, we can derive the graph fractional Laplacian operator of $G_{\square}$, which is $\mathcal{L}_{\alpha}^{prod} = \boldsymbol{\kappa}^{prod}\mathbf{R}^{prod}(\boldsymbol{\kappa}^{prod})^{H}$. For the convenience of subsequent operations, we simply denote it as $\mathcal{L}_{\alpha} = \boldsymbol{\kappa}\mathbf{R}\boldsymbol{\kappa}^{H}$. Thus the input signal is represented as $\mathbf{f} \in \mathbb{R}^{N\times M}$. And the target signal is represented as $\mathbf{g} \in \mathbb{R}^{N}$.

In Section 3, we have already discussed that the graph fractional convolution operator can be expressed as $\mathbf{f} *_{\alpha} \mathbf{h} = \boldsymbol{\kappa}^{H}(\boldsymbol{\kappa}\mathbf{f} \odot \boldsymbol{\kappa}\mathbf{h})$, with $\odot$ denoting a point-wise product. In practical applications, filters are usually designed on a graph in the form of a diagonal matrix $\mathbf{H}(\mathbf{R}) = \mathrm{diag}(\hat{\mathbf{h}})$, where $\hat{\mathbf{h}} = \left[h_1, h_2, \cdots, h_N\right]^{\top}$. Subsequently, the graph fractional convolution operator can be restated as

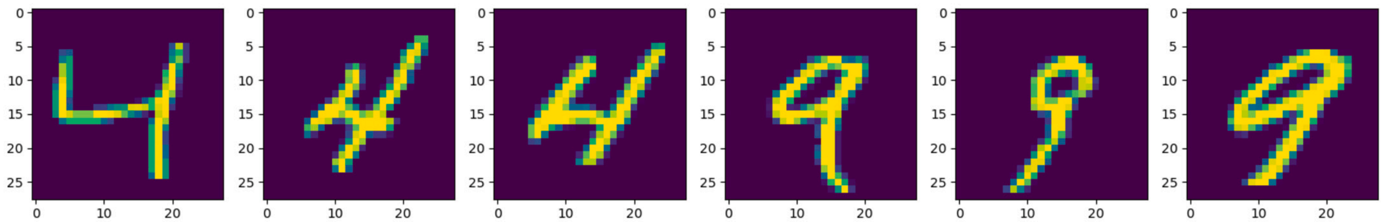


Fig. 3. Samples of MNIST Labeled as 4 and 9.

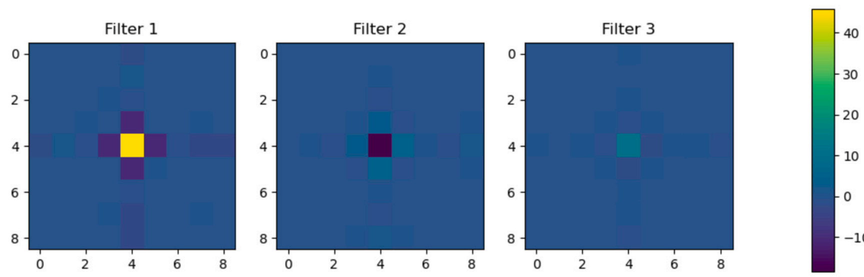


Fig. 4. The Learned Filters $\mathbf{W}_L$ by Using the LanczosNet Method.

$$\mathbf{f} *_\alpha \mathbf{h} = \boldsymbol{\kappa}^H (\boldsymbol{\kappa}\mathbf{f} \odot \boldsymbol{\kappa}\mathbf{h}) = \mathbf{H}(\mathbf{R})\mathbf{f}. \tag{51}$$

In order to enhance the fitting capability of the learner, we additionally introduce a parameterized weight matrix $\mathbf{W}$. Subsequently, the linear mapping can be expressed as

$$\widetilde{\mathbf{g}} = \boldsymbol{\kappa}^H \mathbf{H}(\mathbf{R})\boldsymbol{\kappa}\mathbf{f}\mathbf{W}. \tag{52}$$

Consequently, the selection of an appropriate filter $\mathbf{H}(\mathbf{R})$ plays a crucial role in this image classification task. In this regard, we introduce a filter learning algorithm based on linear least squares to facilitate the learning of the filter.

Algorithm 1: Linear Least Squares Based Filter Learning Algorithm.

Input:
Graph signal: $\mathbf{f}$, Target signal: $\mathbf{g}$.
Initial value of filter: $\mathbf{H}_0$.
Initial value of weight matrix: $\mathbf{W}_0$.
The parameters of fractional order: α.
Regularization parameter: τ, Number of iterations: N_{Iter}.
Output:
The learned filter: $\mathbf{H}_L = \mathrm{diag}(\hat{\mathbf{h}})$.
The learned weight matrix: $\mathbf{W}_L$.
Step 1: Compute $\boldsymbol{\kappa}$.
Step 2: Let $\mathbf{H}_L = \mathbf{H}_0$ and $\mathbf{W}_L = \mathbf{W}_0$.
for $i = 1$ **to** N_{Iter} **do**
Filter update: $\mathbf{H}_L = \left(\boldsymbol{\kappa}\mathbf{f}\mathbf{W}_L\right)^{-1} \cdot (\boldsymbol{\kappa}\mathbf{g})$.
Weight matrix update: $\mathbf{W}_L = \left(\mathbf{H}_L\boldsymbol{\kappa}\mathbf{f} + \tau\right)^{-1} \cdot (\boldsymbol{\kappa}\mathbf{g})$.
end for
Return $\mathbf{H}_L$ and $\mathbf{W}_L$.

Polynomial filters offer the merits of augmenting locality and diminishing the complexity associated with learning [41,42]. A prominent illustration of these filters is LanczosNet, which we employ in the context of our classification endeavor [43].

4.2. Results

Building upon the algorithmic framework presented, we implement the linear least squares based filter learning algorithm for effective sample training. Specifically, we initially employ the algorithm and use the LanczosNet method with an eigenspace dimension of 3. Subsequently, the learned filters $\mathbf{W}_L$ obtained as a result of this optimization procedure are depicted in Fig. 4. In this context, after extensive parameter tuning and empirical validation, we have set the regularization parameter τ to be equal to 10^5, which provides an optimal balance between model stability and learning performance.

Table 1
The Classification Accuracies of Three Methods.

τ	10^4	10^3	10^2	10^{-3}
WGFT	89.29	89.41	89.03	88.99
WGFRFT	89.83	89.29	88.32	89.75
2-D WGFRFT	89.95	90.04	90.02	90.49

Subsequently, we conduct cross-validation on the dataset. We then vary the fractional order parameter α with the aim of observing its influence on the accuracy of the cross-validation results. In this regard, we configure the parameters for cross-validation by setting the number of folds to 10. To illustrate the stability of the model, five independent training runs are carried out, and the average result is adopted as the overall performance evaluation metric. The result is presented in Fig. 5.

When $\alpha = 1$, the 2-D WGFRFT degenerates into the WGFT. It can be noticed that when the fractional order parameter is less than 1, the classification accuracy is higher than that of the WGFT.

The regularization parameter τ constitutes an important parameter within the algorithm. We vary this parameter as well in order to observe its impact on the model. We set $\alpha = 0.25$. The result is presented in Fig. 6. The classification accuracy maintains high performance when τ ranges from 0.01 to 10^5, yet it deteriorates at both ends of this interval.

Under the same conditions, we also test the performance of the WGFT as well as the WGFRFT. When the parameter τ varies, the classification accuracies of the three methods are shown in Table 1. When the range of τ is between 10^{-3} and 10^4, the 2-D WGFRFT achieves the best effect.

Based on the aforementioned experimental validation, our proposed methodology demonstrates considerable efficacy in image classification tasks through the synergistic optimization of two key parameters: the fractional-order parameter α, which controls the transformation characteristics, and the regularization parameter τ, which governs the regularization strength. The judicious tuning of these parameters enables our framework to achieve robust performance across diverse image classification scenarios, thereby substantiating both its adaptability and practical utility.

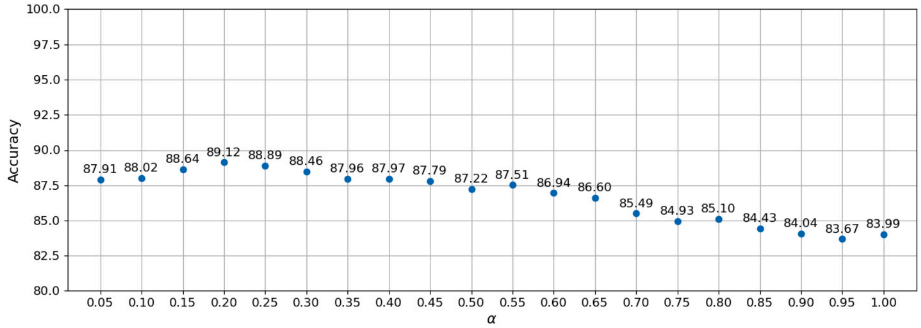


Fig. 5. The Accuracy of the Cross-validation Results by Changing α.

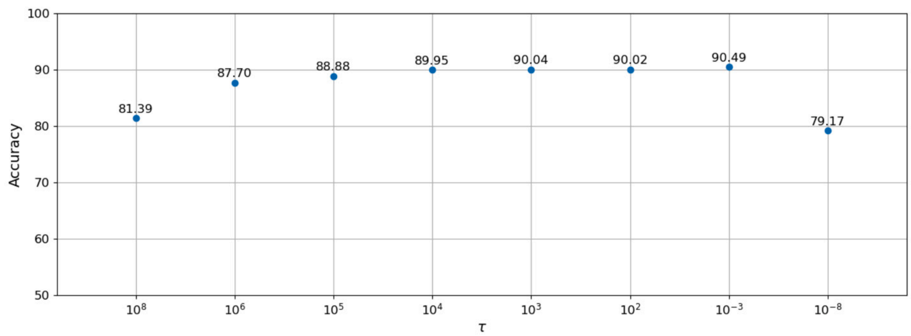


Fig. 6. The Accuracy of the Cross-validation Results by Changing τ.

5. Conclusion

This paper presents the two-dimensional (2-D) windowed graph fractional Fourier transform (WGFRFT) and its fast algorithm. It showcases advantages in handling 2-D graph signals and multi-valued spectral coefficients. The paper illustrates window functions' operation and verifies the fast algorithm's effectiveness. Additionally, the practical application of 2-D WGFRFT is explored. In the paper, a novel filter learning method is proposed based on the 2-D WGFRFT. It shows superior performance, highlighting its potential in practical applications.

CRediT authorship contribution statement

Yu-Chen Gan: original draft and conceptualization. **Jian-Yi Chen:** review and editing. **Bing-Zhao Li:** review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by grants from the National Natural Science Foundation of China [No. 62171041]; and the Key Technologies Special Program of Higher Education in Guangdong Province [No. 2023ZDZX1035]; and the Natural Science Foundation of Beijing Municipality [No. 4242011].

Appendix A. The proof of the I-2-D WGFRFT

Proof. It can be seen from Definition 3 that

$$\sum_{i_1=1}^{N_1}\sum_{i_2=1}^{N_2}\sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1,l_2)\boldsymbol{\Psi}_{i_1,i_2,l_1,l_2}(n)$$

$$\begin{aligned}
&=\sum_{i_1=1}^{N_1}\sum_{i_2=1}^{N_2}\sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}((N_1N_2)^{\alpha}\sum_{d_1=1}^{N_1}\sum_{d_2=1}^{N_2}f(d_1,d_2)\\
&\quad\times\left(\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\widehat{\psi}_{\alpha}^{*}(m_1,m_2)\kappa_{m_1}^{(1)}(i_1)\kappa_{m_2}^{(2)}(i_2)\kappa_{m_1}^{(1)*}(d_1)\kappa_{m_2}^{(2)*}(d_2)\right)\\
&\quad\times\kappa_{l_1}^{(1)*}(d_1)\kappa_{l_2}^{(2)*}(d_2))\\
&\quad\times\Bigg((N_1N_2)^{\alpha}\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2)\sum_{j_1=1}^{N_1}\sum_{j_2=1}^{N_2}\widehat{\psi}_{\alpha}(j_1,j_2)\kappa_{j_1}^{(1)*}(i_1)\kappa_{j_2}^{(2)*}(i_2)\\
&\quad\times\kappa_{j_1}^{(1)}(n_1)\kappa_{j_2}^{(2)}(n_2)\Bigg)\\
&=(N_1N_2)^{2\alpha}\sum_{d_1=1}^{N_1}\sum_{d_2=1}^{N_2}f(d_1,d_2)\\
&\quad\times\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\sum_{j_1=1}^{N_1}\sum_{j_2=1}^{N_2}\widehat{\psi}_{\alpha}^{*}(m_1,m_2)\widehat{\psi}_{\alpha}(j_1,j_2)\kappa_{m_1}^{(1)*}(d_1)\kappa_{m_2}^{(2)*}(d_2)\\
&\quad\times\kappa_{j_1}^{(1)}(n_1)\kappa_{j_2}^{(2)}(n_2)\\
&\quad\times\sum_{i_1=1}^{N_1}\sum_{i_2=1}^{N_2}\kappa_{m_1}^{(1)}(i_1)\kappa_{m_2}^{(2)}(i_2)\kappa_{j_1}^{(1)*}(i_1)\kappa_{j_2}^{(2)*}(i_2)\\
&\quad\times\sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\kappa_{l_1}^{(1)*}(d_1)\kappa_{l_2}^{(2)*}(d_2)\kappa_{l_1}^{(1)}(n_1)\kappa_{l_2}^{(2)}(n_2)\\
&=(N_1N_2)^{2\alpha}\sum_{d_1=1}^{N_1}\sum_{d_2=1}^{N_2}f(d_1,d_2)\\
&\quad\times\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\sum_{j_1=1}^{N_1}\sum_{j_2=1}^{N_2}\widehat{\psi}_{\alpha}^{*}(m_1,m_2)\widehat{\psi}_{\alpha}(j_1,j_2)\kappa_{m_1}^{(1)*}(d_1)\kappa_{m_2}^{(2)*}(d_2)\kappa_{j_1}^{(1)}(n_1)\\
&\quad\times\kappa_{j_2}^{(2)}(n_2)\delta_{(m_1,m_2)(j_1,j_2)}\delta_{(d_1,d_2)(n_1,n_2)}
\end{aligned}$$

$$=(N_1N_2)^{2\alpha}f(n_1,n_2)\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\left|\widehat{\varphi}_\alpha(m_1,m_2)\right|^2\left|\kappa_{m_1}^{(1)*}(n_1)\kappa_{m_2}^{(2)*}(n_2)\right|^2. \quad \text{(A.1)}$$

Moreover, according to Definition 1, it can be found that

$$\begin{aligned}
&\left\|T_{i_1,i_2}^{\alpha}\psi\right\|_2^2\\
&=\sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2}\Bigg((\sqrt{N_1N_2})^{\alpha}\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\widehat{\varphi}_\alpha(m_1,m_2)\kappa_{m_1}^{(1)*}(i_1)\kappa_{m_2}^{(2)*}(i_2)\\
&\times\kappa_{m_1}^{(1)}(n_1)\kappa_{m_2}^{(2)}(n_2)\Bigg)^2\\
&=(N_1N_2)^{\alpha}\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\sum_{m_1'=1}^{N_1}\sum_{m_2'=1}^{N_2}\widehat{\varphi}_\alpha(m_1,m_2)\widehat{\varphi}_\alpha(m_1',m_2')\kappa_{m_1}^{(1)*}(i_1)\kappa_{m_2}^{(2)*}(i_2)\\
&\times\kappa_{m_1'}^{(1)*}(i_1)\kappa_{m_2'}^{(2)*}(i_2)\\
&\times\sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2}\kappa_{m_1}^{(1)}(n_1)\kappa_{m_2}^{(2)}(n_2)\kappa_{m_1'}^{(1)}(n_1)\kappa_{m_2'}^{(2)}(n_2)\\
&=(N_1N_2)^{\alpha}f(n_1,n_2)\sum_{m_1=1}^{N_1}\sum_{m_2=1}^{N_2}\left|\widehat{\varphi}_\alpha(m_1,m_2)\right|^2\left|\kappa_{m_1}^{(1)*}(n_1)\kappa_{m_2}^{(2)*}(n_2)\right|^2.
\end{aligned} \quad \text{(A.2)}$$

Thus, it follows that

$$\begin{aligned}
&\sum_{i_1=1}^{N_1}\sum_{i_2=1}^{N_2}\sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1,l_2)\boldsymbol{\Psi}_{i_1,i_2,l_1,l_2}(n)\\
&=(N_1N_2)^{\alpha}f(n_1,n_2)\left\|T_{i_1,i_2}^{\alpha}\psi\right\|_2^2.
\end{aligned} \quad \text{(A.3)}$$

Therefore, there is

$$\begin{aligned}
f(n_1,n_2)=&\frac{1}{(N_1N_2)^{\alpha}\left\|T_{i_1,i_2}^{\alpha}\boldsymbol{\psi}\right\|_2^2}\sum_{i_1=1}^{N_1}\\
&\times\sum_{i_2=1}^{N_2}\sum_{l_1=1}^{N_1}\sum_{l_2=1}^{N_2}\left(W_{\psi,\alpha}\mathbf{f}\right)(i_1,i_2,l_1,l_2)\boldsymbol{\Psi}_{i_1,i_2,l_1,l_2}(n_1,n_2). \qquad \square
\end{aligned} \quad \text{(A.4)}$$

Data availability

No data was used for the research described in the article.

References

[1] F. Gama, E. Isufi, G. Leus, A. Ribeiro, Graphs, convolutions, and neural networks: from graph filters to graph neural networks, IEEE Signal Process. Mag. 37 (6) (2020) 128–138.

[2] M.T. Schaub, Y. Zhu, J.-B. Seby, T.M. Roddenberry, S. Segarra, Signal processing on higher-order networks: Livin' on the edge... and beyond, Signal Process. 187 (2021) 108149.

[3] K. Liao, Z. Yu, N. Xie, J. Jiang, Joint estimation of azimuth and distance for far-field multi targets based on graph signal processing, Remote Sens. 14 (5) (2022) 1110.

[4] M. Petrovic, R. Liegeois, T.A. Bolton, D. Van De Ville, Community-aware graph signal processing: modularity defines new ways of processing graph signals, IEEE Signal Process. Mag. 37 (6) (2020) 150–159.

[5] R. Li, X. Yuan, M. Radfar, P. Marendy, W. Ni, T.J. O'Brien, P.M. Casillas-Espinosa, Graph signal processing, graph neural network and graph learning on biological data: a systematic review, IEEE Rev. Biomed. Eng. 16 (2021) 109–135.

[6] S.S. Saboksayr, G. Mateos, M. Cetin, Eeg-based emotion classification using graph signal processing, in: ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2021, pp. 1065–1069.

[7] A. Sandryhaila, J.M. Moura, Big data analysis with signal processing on graphs: representation and processing of massive data sets with irregular structure, IEEE Signal Process. Mag. 31 (5) (2014) 80–90.

[8] A. Ortega, P. Frossard, J. Kovačević, J.M. Moura, P. Vandergheynst, Graph signal processing: overview, challenges, and applications, Proc. IEEE 106 (5) (2018) 808–828.

[9] G. Leus, A.G. Marques, J.M. Moura, A. Ortega, D.I. Shuman, Graph signal processing: history, development, impact, and outlook, IEEE Signal Process. Mag. 40 (4) (2023) 49–60.

[10] R. Ramakrishna, H.-T. Wai, A. Scaglione, A user guide to low-pass graph signal processing and its applications: tools and applications, IEEE Signal Process. Mag. 37 (6) (2020) 74–85.

[11] A. Sandryhaila, J.M. Moura, Discrete signal processing on graphs: graph Fourier transform, in: 2013 IEEE International Conference on Acoustics, Speech and Signal Processing, IEEE, 2013, pp. 6167–6170.

[12] S. Chen, R. Varma, A. Sandryhaila, J. Kovačević, Discrete signal processing on graphs: sampling theory, IEEE Trans. Signal Process. 63 (24) (2015) 6510–6523.

[13] A. Mazarguil, L. Oudre, N. Vayatis, Localized interpolation for graph signals, in: 2020 28th European Signal Processing Conference (EUSIPCO), IEEE, 2021, pp. 2160–2164.

[14] J.Y. Chen, Y. Zhang, B.Z. Li, Graph linear canonical transform: definition, vertex-frequency analysis and filter design, IEEE Trans. Signal Process. 72 (2024) 5691–5707, https://doi.org/10.1109/TSP.2024.3507787.

[15] D. Wei, Z. Yan, Generalized sampling of graph signals with the prior information based on graph fractional Fourier transform, Signal Process. 214 (2024) 109263.

[16] D.I. Shuman, S.K. Narang, P. Frossard, A. Ortega, P. Vandergheynst, The emerging field of signal processing on graphs: extending high-dimensional data analysis to networks and other irregular domains, IEEE Signal Process. Mag. 30 (3) (2013) 83–98.

[17] L.B. Almeida, The fractional Fourier transform and time-frequency representations, IEEE Trans. Signal Process. 42 (11) (1994) 3084–3091.

[18] H.M. Ozaktas, O. Arikan, M.A. Kutay, G. Bozdagt, Digital computation of the fractional Fourier transform, IEEE Trans. Signal Process. 44 (9) (1996) 2141–2150.

[19] Y.-q. Wang, B.-z. Li, Q.-y. Cheng, The fractional Fourier transform on graphs, in: 2017 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE, 2017, pp. 105–110.

[20] J. Wu, F. Wu, Q. Yang, Y. Zhang, X. Liu, Y. Kong, L. Senhadji, H. Shu, Fractional spectral graph wavelets and their applications, Math. Probl. Eng. 2020 (1) (2020) 2568179.

[21] W. Imrich, S. Klavzar, D.F. Rall, Topics in Graph Theory: Graphs and Their Cartesian Product, CRC Press, 2008.

[22] T. Kurokawa, T. Oki, H. Nagao, Multi-dimensional graph Fourier transform, arXiv preprint, arXiv:1712.07811.

[23] A. Natali, E. Isufi, G. Leus, Forecasting multi-dimensional processes over graphs, in: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 5575–5579.

[24] R.A. Varma, J. Kovacevic, Sampling theory for graph signals on product graphs, in: 2018 IEEE Global Conference on Signal and Information Processing (GlobalSIP), IEEE, 2018, pp. 768–772.

[25] J.K. Gautam, A. Kumar, R. Saxena, Windows: a tool in signal processing, IETE Tech. Rev. 12 (3) (1995) 217–226.

[26] K.M. Chew, R. Sudirman, N. Seman, C.Y. Yong, Signal processing of microwave imaging brain tumor detection using superposition windowing, Appl. Mech. Mater. 654 (2014) 321–326.

[27] H. Kaplan, A. Nacaroğlu, A new window function for fir filter design, in: 2015 23rd Signal Processing and Communications Applications Conference (SIU), IEEE, 2015, pp. 2458–2461.

[28] P. Agarwal, S. Kumar, S. Singh, Closed form solutions of various window functions in fractional Fourier transform domain, in: 2019 6th International Conference on Computing for Sustainable Global Development (INDIACom), IEEE, 2019, pp. 64–68.

[29] A.K. Singh, V.K. Srivastava, Performance evaluation of different window functions for stdft based exon prediction technique taking paired numeric mapping scheme, in: 2019 6th International Conference on Signal Processing and Integrated Networks (SPIN), IEEE, 2019, pp. 739–743.

[30] F. Albasu, M. Kulyabin, A. Zhdanov, A. Dolganov, M. Ronkin, V. Borisov, L. Dorosinsky, P.A. Constable, M.A. Al-Masni, A. Maier, Electroretinogram analysis using a short-time Fourier transform and machine learning techniques, Bioengineering 11 (9) (2024) 866.

[31] D.I. Shuman, B. Ricaud, P. Vandergheynst, A windowed graph Fourier transform, in: 2012 IEEE Statistical Signal Processing Workshop (SSP), Ieee, 2012, pp. 133–136.

[32] F.-J. Yan, B.-Z. Li, Windowed fractional Fourier transform on graphs: properties and fast algorithm, Digit. Signal Process. 118 (2021) 103210.

[33] F.-J. Yan, W.-B. Gao, B.-Z. Li, Windowed fractional Fourier transform on graphs: fractional translation operator and Hausdorff-Young inequality, in: 2020 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE, 2020, pp. 255–259.

[34] C.-C. Tseng, S.-L. Lee, Computation of graph Fourier transform centrality using graph filter, IEEE Open J. Circuits Syst. 5 (2024) 69–80.

[35] F.-J. Yan, B.-Z. Li, Multi-dimensional graph fractional Fourier transform and its application to data compression, Digit. Signal Process. 129 (2022) 103683.

[36] S.K. Kadambari, S.P. Chepuri, Learning product graphs from multidomain signals, in: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 5665–5669.

[37] R. Tao, Y.-L. Li, Y. Wang, Short-time fractional Fourier transform and its applications, IEEE Trans. Signal Process. 58 (5) (2009) 2568–2580.

[38] A. Baldominos, Y. Saez, P. Isasi, A survey of handwritten character recognition with mnist and emnist, Appl. Sci. 9 (15) (2019) 3169.

[39] L. Deng, The mnist database of handwritten digit images for machine learning research [best of the web], IEEE Signal Process. Mag. 29 (6) (2012) 141–142.

[40] S.S. Kadam, A.C. Adamuthe, A.B. Patil, Cnn model for image classification on mnist and fashion-mnist dataset, J. Sci. Res. 64 (2) (2020) 374–384.

[41] V.J. Mathews, Adaptive polynomial filters, IEEE Signal Process. Mag. 8 (3) (1991) 10–26.

[42] M. Defferrard, X. Bresson, P. Vandergheynst, Convolutional neural networks on graphs with fast localized spectral filtering, Advances in Neural Information Processing Systems 29.

[43] R. Liao, Z. Zhao, R. Urtasun, R.S. Zemel, Lanczosnet: multi-scale deep graph convolutional networks, arXiv preprint, arXiv:1901.01484.

Yu-Chen Gan was born in Jiangxi, China in 2002. She graduated from Beijing Institute of Technology, Beijing, China, with a Bachelor of Science degree. Currently, she is pursuing a Master of Science degree at the same prestigious institution. Gan's research predominantly focuses on the graph Fourier transform and graph fractional Fourier transform.

Jian-Yi Chen was born in Zhuozhou, China, in 1998. She received the M.S. degree from Beijing Institute of Technology, Beijing, China. She is currently working towards the Ph.D. degree in Beijing Institute of Technology, Beijing, China. Her research interests include the theory and applications of linear canonical transform, analytic signal and time-frequency analysis.

Bing-Zhao Li was born in 1975. He received the B.S. degree from Shandong Normal University in 1998 and the MS. and Ph.D. degrees from the Beijing Institute of Technology, in 2001 and 2007, respectively. He is currently a Professor with the School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, China. His current research interests include the theory and applications of fractional Fourier transform, linear canonical transform, and fractional calculus. Author Biography.